Experiment concepts in description logic

with a partonomy, and the challenge-rescue experiment as an instance

Generated from `experiment_concepts.ldl` by `rsa/experiment_dl.py`

Conventions

The concepts are stated in $\mathcal{SROIQ}(\mathbf{D})$, the logic of OWL 2 DL. Concept names are set in sans, role names in *italics*, individuals in mono. A filler restriction $(=n\ r.C)$ means exactly n r-successors in C; $\exists r.C$ at least one; $\forall r.C$ only C. Each LDL clause `(has role pattern cardinality)` becomes one number restriction, `(only role pattern)` one value restriction, a pattern's slots nested existentials, and `(role x)` the restriction $\exists \mathit{hasRole}.\mathsf{X}\ \mathsf{Role}$.

Conformance of an experiment is checked under the *closed-world* reading: the axioms are integrity constraints over the instance as given, derived and placed (`rsa/experiment_concepts.py`). An OWL reasoner reads the same axioms under the open world: it would infer a missing culture rather than report it. Constraints that $\mathcal{SROIQ}(\mathbf{D})$ cannot express are stated in Section 6 in the logic they need.

1 Roles and partonomy (RBox)

Parthood (*hasPart*) is transitive and has four kinds: temporal parts of processes (*hasPhase*, *hasStep*), material parts (*hasWell*) and components of plans and information entities (*hasComponent*: *hasExposure*, *hasObservation*, *hasCandidate*). Three relations are deliberately *not* parthood: a culture is *contained in* a well (location), an action *realizes* a part of the plan, and a culture *participates* in the actions it undergoes.

$\mathit{hasPhase} \sqsubseteq \mathit{hasPart}$

$\mathit{hasStep} \sqsubseteq \mathit{hasPart}$

$\mathit{hasComponent} \sqsubseteq \mathit{hasPart}$

$\mathit{hasExposure} \sqsubseteq \mathit{hasComponent}$

$\mathit{hasObservation} \sqsubseteq \mathit{hasComponent}$

$\mathit{hasCandidate} \sqsubseteq \mathit{hasComponent}$

$\mathit{hasWell} \sqsubseteq \mathit{hasPart}$

$\mathit{hasUnit} \sqsubseteq \mathit{hasParticipant}$

$\mathit{hasCulture} \sqsubseteq \mathit{hasUnit}$

$\mathit{hasSubstance} \sqsubseteq \mathit{hasParticipant}$

$\mathit{hasSite} \sqsubseteq \mathit{hasParticipant}$

$\mathit{hasQuestion} \sqsubseteq \mathit{specifies}$

$\mathit{hasGoal} \sqsubseteq \mathit{specifies}$

$\mathit{hasSharedConditions} \sqsubseteq \mathit{specifies}$

$\mathit{hasComparison} \sqsubseteq \mathit{specifies}$

$\mathit{prevents} \sqsubseteq \mathit{specifies}$

$\mathit{maintains} \sqsubseteq \mathit{specifies}$

$\mathit{hasInsight} \sqsubseteq \mathit{hasOutput}$

$\mathit{Trans}(\mathit{hasPart})$

$\mathit{partOf} \equiv \mathit{hasPart}^{-}$

$\mathit{containedIn} \equiv \mathit{contains}^{-}$

$\mathit{containedIn} \circ \mathit{partOf} \sqsubseteq \mathit{locatedIn}$

$\mathit{hasPart} \circ \mathit{hasParticipant} \sqsubseteq \mathit{hasParticipant}$

OWL 2 allows number restrictions only on simple roles. Non-simple here: *hasPart*, *hasParticipant*, *locatedIn*, *partOf*. Roles in number restrictions: *hasBelief*, *hasCandidate*, *hasComparison*, *contains*, *hasGoal*, *hasInsight*, *hasPhase*, *hasQuestion*, *hasSharedConditions*, *hasSite*, *hasSubstance*, *hasUnit*. All are simple.

2 Kinds and their partonomy

Four disjoint kinds. Parthood never crosses kinds: a process has processes as parts, a plate has wells, a treatment has exposures, a question has candidate hypotheses.

Process ⊓ MaterialEntity ⊑ ⊥

Process ⊓ PlanSpecification ⊑ ⊥

Process ⊓ InformationEntity ⊑ ⊥

MaterialEntity ⊓ PlanSpecification ⊑ ⊥

MaterialEntity ⊓ InformationEntity ⊑ ⊥

PlanSpecification ⊓ InformationEntity ⊑ ⊥

Process ⊑ ∀*hasPart*.Process

MaterialEntity ⊑ ∀*hasPart*.MaterialEntity

PlanSpecification ⊑ ∀*hasPart*.PlanSpecification

InformationEntity ⊑ ∀*hasPart*.InformationEntity

3 Entities

Action ⊑ Process

Culture ⊑ MaterialEntity

Substance ⊑ MaterialEntity

Plate ⊑ MaterialEntity

Plate ⊑ ∃*hasWell*.Well

Well ⊑ MaterialEntity

Well ⊑ (≤ 1*contains*.⊤)

ControlledEnvironment ⊑ MaterialEntity

Treatment ⊑ PlanSpecification

Treatment ⊑ ∀*hasExposure*.(Exposure ⊓ ∃*exposedTo*.⊤ ⊓ ∃*dose*.⊤ ⊓ ∃*atTime*.⊤)

Exposure ⊑ PlanSpecification

Condition ⊑ PlanSpecification

Observation ⊑ PlanSpecification

Estimate ⊑ InformationEntity

Scale ⊑ InformationEntity

Image ⊑ InformationEntity

Comparison ⊑ PlanSpecification

Question ⊑ InformationEntity

RecoveryModel ⊑ InformationEntity

Belief ⊑ InformationEntity

Posterior ⊑ InformationEntity

Insight ⊑ InformationEntity

Effect ⊑ InformationEntity

InterventionAction ⊑ Action

GrowthAction ⊑ Action

ObservationAction ⊑ Action

InterpretationAction ⊑ Action

AnalysisAction ⊑ Action

EmptyTreatment ≡ Treatment ⊓ ¬∃*hasExposure*.⊤

4 Phases: the process partonomy of an experiment

Phase — a temporal part of an experiment.

Phase ⊑ Process

Phase ⊑ ∃*hasStep*.Action

CultivationPhase — the units are treated and kept under the shared conditions.

CultivationPhase ⊑ Phase

CultivationPhase ⊑ ∀*hasStep*.(InterventionAction ⊔ GrowthAction)

ObservationPhase — the units are observed.

ObservationPhase ⊑ Phase

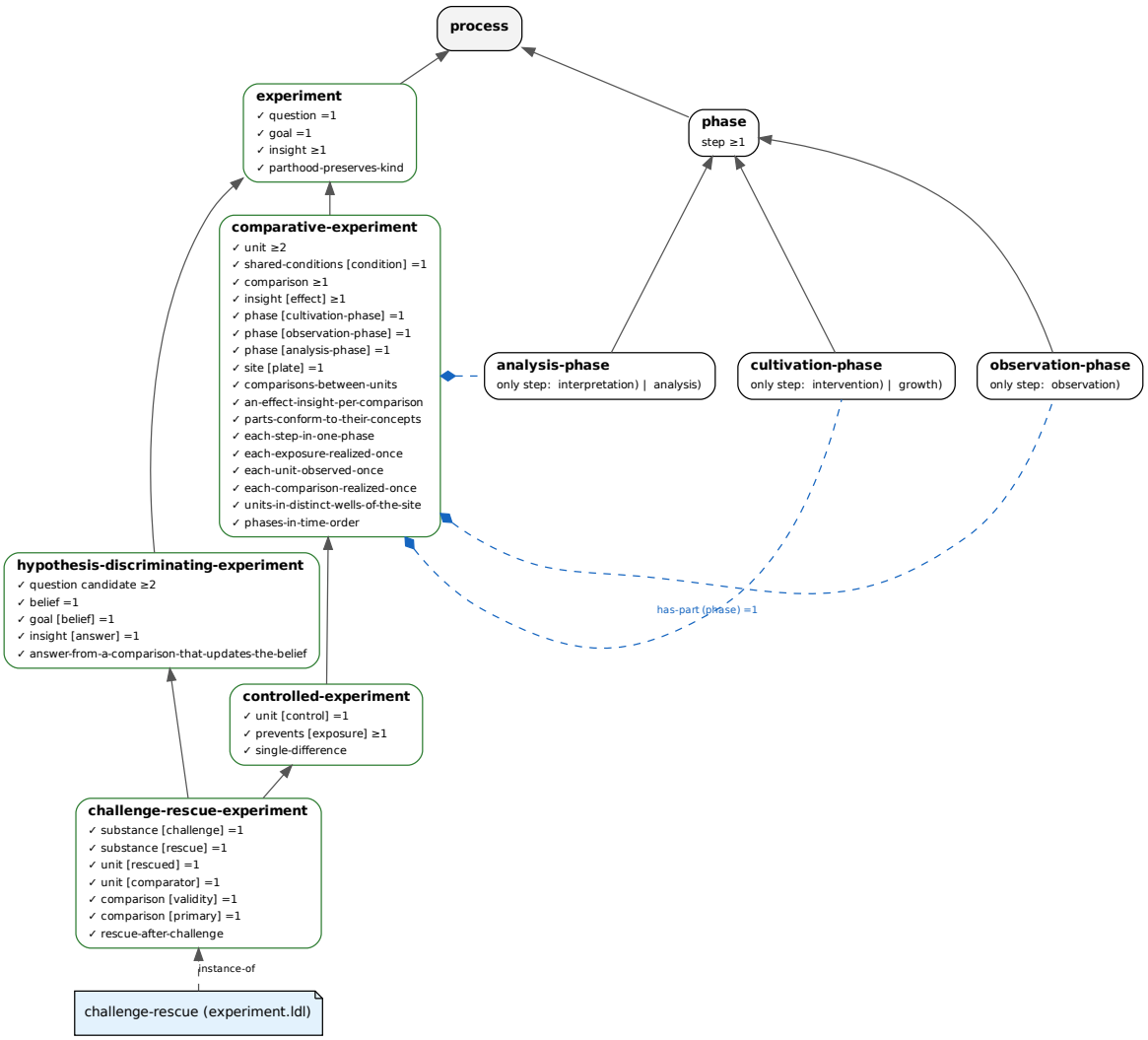
ObservationPhase ⊑ ∀*hasStep*.ObservationAction

AnalysisPhase — the observations are interpreted and compared.

AnalysisPhase ⊑ Phase

AnalysisPhase ⊑ ∀*hasStep*.(InterpretationAction ⊔ AnalysisAction)

5 Experiments



Experiment

an activity that answers a question by producing insights.

Experiment $\sqsubseteq$ Process

Experiment $\sqsubseteq (= 1 \mathit{hasQuestion}.\text{Question})$

Experiment $\sqsubseteq (= 1 \mathit{hasGoal}.\top)$

Experiment $\sqsubseteq \exists \mathit{hasInsight}.\text{Insight}$

Further constraints (Section 6): `parthood-preserves-kind`.

HypothesisDiscriminatingExperiment

the question asks which of several candidate hypotheses holds; the experiment changes a belief over them, and its answer is the posterior.

HypothesisDiscriminatingExperiment $\sqsubseteq$ Experiment

HypothesisDiscriminatingExperiment $\sqsubseteq \exists \mathit{hasQuestion}.(\geq 2 \mathit{hasCandidate}.\top)$

HypothesisDiscriminatingExperiment $\sqsubseteq (= 1 \mathit{hasBelief}.(\text{Belief} \sqcap \exists \mathit{over}.\top \sqcap \exists \mathit{hasPrior}.\top))$

HypothesisDiscriminatingExperiment $\sqsubseteq (= 1\,\mathit{hasGoal}.(\text{Belief} \sqcap \exists \mathit{expectedInformationGain}.\top))$

HypothesisDiscriminatingExperiment $\sqsubseteq (= 1\,\mathit{hasInsight}.(\text{Insight} \sqcap \exists \mathit{hasRole}.\text{AnswerRole} \sqcap \exists \mathit{hasForm}.\text{Posterior}))$

Further constraints (Section 6): `answer-from-a-comparison-that-updates-the-belief`.

ComparativeExperiment

units are treated differently, kept under shared conditions and observed the same way; comparisons between units determine the effects of the treatments. Its procedure yields three phases.

ComparativeExperiment $\sqsubseteq$ Experiment

ComparativeExperiment $\sqsubseteq (\geq 2\,\mathit{hasUnit}.\exists \mathit{hasTreatment}.\top)$

ComparativeExperiment $\sqsubseteq (= 1\,\mathit{hasSharedConditions}.(\text{Condition} \sqcap \exists \mathit{fromTime}.\top \sqcap \exists \mathit{toTime}.\top \sqcap \exists \mathit{hasObservation}.(\text{Observation} \sqcap \exists \mathit{atTime}.\top \sqcap \exists \mathit{interpretedAs}.\top)))$

ComparativeExperiment $\sqsubseteq \exists \mathit{hasComparison}.(\text{Comparison} \sqcap \exists \mathit{between}.\top \sqcap \exists \mathit{determines}.\text{Effect})$

ComparativeExperiment $\sqsubseteq \exists \mathit{hasInsight}.(\text{Insight} \sqcap \exists \mathit{hasRole}.\text{EffectRole})$

ComparativeExperiment $\sqsubseteq (= 1\,\mathit{hasPhase}.\text{CultivationPhase})$

ComparativeExperiment $\sqsubseteq (= 1\,\mathit{hasPhase}.\text{ObservationPhase})$

ComparativeExperiment $\sqsubseteq (= 1\,\mathit{hasPhase}.\text{AnalysisPhase})$

ComparativeExperiment $\sqsubseteq (= 1\,\mathit{hasSite}.\text{Plate})$

Further constraints (Section 6): `comparisons-between-units`, `an-effect-insight-per-comparison`, `parts-conform-to-their-concepts`, `each-step-in-one-phase`, `each-exposure-realized-once`, `each-unit-observed-once`, `each-comparison-realized-once`, `units-in-distinct-wells-of-the-site`, `phases-in-time-order`. Procedure (rules that derive the actions; not DL axioms):

- `(rule (each exposure (of (a unit))) (yields (an action (kind intervention) (in (the cultivation-phase)))))`
- `(rule (each interval (between (consecutive treatment-times))) (yields (an action (kind growth) (in (the cultivation-phase)))))`
- `(rule (each unit) (yields (an action (kind observation) (in (the observation-phase))) (an action (kind interpretation) (in (the analysis-phase)))))`
- `(rule (each comparison) (yields (an action (kind analysis) (in (the analysis-phase)))))`
- `(rule (a comparison (licenses ?c)) (orders (its analysis) (before (the analysis (of ?c)))))`

ControlledExperiment

one unit is an untreated control, it stays untreated, and every comparison isolates exactly one exposure: its two units differ in that exposure only, and it is the exposure whose effect the comparison determines.

ControlledExperiment $\sqsubseteq$ ComparativeExperiment

ControlledExperiment $\sqsubseteq (= 1\,\mathit{hasUnit}.(\exists \mathit{hasRole}.\text{ControlRole} \sqcap \exists \mathit{hasTreatment}.\text{EmptyTreatment}))$

ControlledExperiment $\sqsubseteq \exists \mathit{prevents}.(\text{Exposure} \sqcap \exists \mathit{of}.(\text{Culture} \sqcap \exists \mathit{hasRole}.\text{ControlRole}))$

Further constraints (Section 6): `single-difference`.

ChallengeRescueExperiment

units are injured by a challenge; one is then given a rescue after a delay. A validity comparison (comparator vs control) establishes the challenge's effect and licenses the primary comparison (rescued vs comparator), which determines the rescue's effect and updates the belief over models of recovery.

ChallengeRescueExperiment $\sqsubseteq$ ControlledExperiment $\sqcap$ HypothesisDiscriminatingExperiment

ChallengeRescueExperiment $\sqsubseteq (= 1\,\mathit{hasSubstance}.(\text{Substance} \sqcap \exists \mathit{hasRole}.\text{ChallengeRole}))$

ChallengeRescueExperiment $\sqsubseteq (= 1\,\mathit{hasSubstance}.(\text{Substance} \sqcap \exists \mathit{hasRole}.\text{RescueRole}))$

ChallengeRescueExperiment $\sqsubseteq$ ($=$ 1$hasUnit$.(Culture $\sqcap$ $\exists hasRole$.RescuedRole $\sqcap$ $\exists hasTreatment$.(Treatment $\sqcap$ $\exists hasExposure$.(Exposure $\sqcap$ $\exists exposedTo$.(Substance $\sqcap$ $\exists hasRole$.ChallengeRole)) $\sqcap$ $\exists hasExposure$.(Exposure $\sqcap$ $\exists exposedTo$.(Substance $\sqcap$ $\exists hasRole$.RescueRole)))))

ChallengeRescueExperiment $\sqsubseteq$ ($=$ 1$hasUnit$.(Culture $\sqcap$ $\exists hasRole$.ComparatorRole $\sqcap$ $\exists hasTreatment$.(Treatment $\sqcap$ $\exists hasExposure$.(Exposure $\sqcap$ $\exists exposedTo$.(Substance $\sqcap$ $\exists hasRole$.ChallengeRole)))))

ChallengeRescueExperiment $\sqsubseteq$ ($=$ 1$hasComparison$.(Comparison $\sqcap$ $\exists hasRole$.ValidityRole $\sqcap$ $\exists between$.(Culture $\sqcap$ $\exists hasRole$.ComparatorRole) $\sqcap$ $\exists between$.(Culture $\sqcap$ $\exists hasRole$.ControlRole) $\sqcap$ $\exists determines$.(Effect $\sqcap$ $\exists of$.(Substance $\sqcap$ $\exists hasRole$.ChallengeRole)) $\sqcap$ $\exists licenses$.(Comparison $\sqcap$ $\exists hasRole$.PrimaryRole)))

ChallengeRescueExperiment $\sqsubseteq$ ($=$ 1$hasComparison$.(Comparison $\sqcap$ $\exists hasRole$.PrimaryRole $\sqcap$ $\exists between$.(Culture $\sqcap$ $\exists hasRole$.RescuedRole) $\sqcap$ $\exists between$.(Culture $\sqcap$ $\exists hasRole$.ComparatorRole) $\sqcap$ $\exists determines$.(Effect $\sqcap$ $\exists of$.(Substance $\sqcap$ $\exists hasRole$.RescueRole)) $\sqcap$ $\exists updates$.Belief))

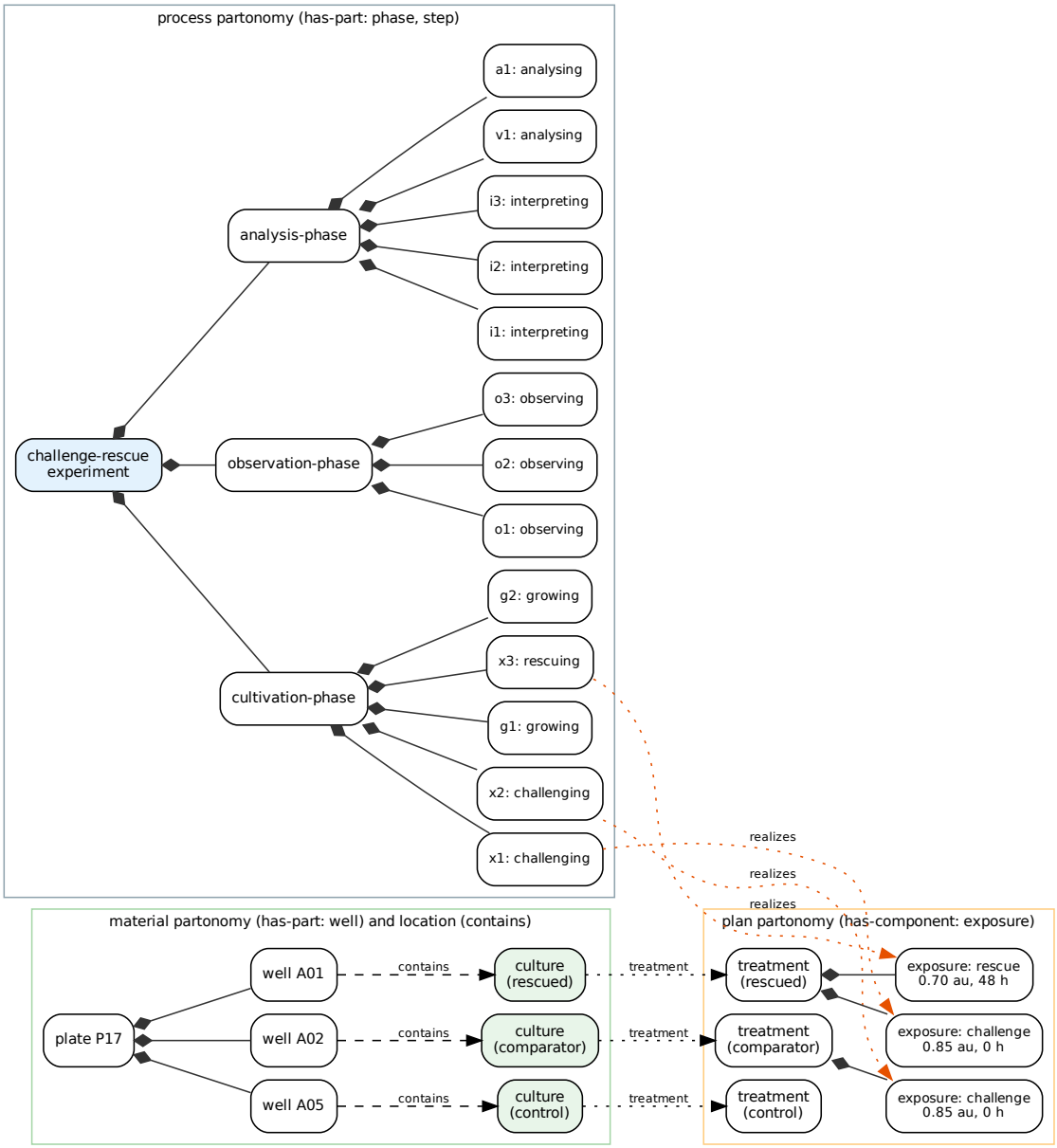
Further constraints (Section 6): `rescue-after-challenge`.

6 Constraints beyond the TBox axioms

constraint	logic	statement
`parthood-preserves-kind`	OWL 2 DL	Process $\sqsubseteq \forall hasPart$.Process MaterialEntity $\sqsubseteq \forall hasPart$.MaterialEntity PlanSpecification $\sqsubseteq \forall hasPart$.PlanSpecification InformationEntity $\sqsubseteq \forall hasPart$.InformationEntity *stated in the kinds section; parthood never crosses kinds*
`answer-from-a-comparison-that-updates-the-belief`	OWL 2 DL	Insight $\sqcap \exists hasRole$.AnswerRole $\sqsubseteq \exists from.$(Comparison $\sqcap \exists updates$.Belief) *that the comparison is one of the same experiment needs variable sharing (FOL)*
`comparisons-between-units`	SROIQ role inclusion	$hasComparison \circ between \sqsubseteq hasUnit$ *read as an integrity constraint: under the open world an OWL reasoner would infer the unit, not report it*
`an-effect-insight-per-comparison`	FOL	$\forall e, c.hasComparison(e, c) \rightarrow \exists i.hasInsight(e, i) \wedge from(i, c) \wedge$ $\forall s.(determines \circ of)(c, s) \rightarrow of(i, s)$ *OWL 2 states only* Comparison $\sqsubseteq \exists from^{-}$.(Insight $\sqcap \exists hasRole$.EffectRole)
`parts-conform-to-their-concepts`	DL semantics	(every TBox axiom holds of every individual) *the closed-world checker evaluates each part against its own concept*
`each-step-in-one-phase`	OWL 2 DL	Action $\sqsubseteq (= 1 hasStep^{-}$.Phase)
`each-exposure-realized-once`	OWL 2 DL + rule	Exposure $\sqsubseteq (= 1 realizes^{-}$.InterventionAction) $realizes(a, x) \leftarrow hasCulture(a, u) \wedge hasTreatment(u, t) \wedge$ $hasExposure(t, x) \wedge$ same substance, dose and time *the rule that computes realizes compares data values of two individuals (SWRL)*
`each-unit-observed-once`	FOL	$\forall e, u.hasUnit(e, u) \rightarrow \exists^{=1} s.hasPart(e, s) \wedge$ ObservationAction$(s) \wedge$ $hasCulture(s, u)$ *OWL 2 states only* Culture $\sqsubseteq (= 1 hasCulture^{-}$.ObservationAction*), not 'within the same experiment'*
`each-comparison-realized-once`	OWL 2 DL + rule	Comparison $\sqsubseteq (= 1 realizes^{-}$.AnalysisAction) *realizes is computed from the analysis's statistic (the pair of units it compares)*
`units-in-distinct-wells-of-the-site`	FOL (path agreement)	Culture $\sqsubseteq (= 1 containedIn$.Well), Well $\sqsubseteq (\leq 1 contains.\top)$ $\forall e, u, v.hasUnit(e, u) \wedge$ $hasUnit(e, v) \rightarrow (containedIn \circ partOf)(u) = (containedIn \circ partOf)(v)$ *the first line is OWL 2 DL; 'one plate for all units' is a role-value map, undecidable with the rest*
`phases-in-time-order`	ALC(D) feature chains	ComparativeExperiment $\sqsubseteq$ $\exists(cultivation \circ toTime, observationPhase \circ atTime). \leq$ *compares two data values reached by different paths: a concrete-domain predicate, not available in OWL 2*
`single-difference`	FOL with counting	$\forall c, u, v.between(c, u) \wedge between(c, v) \wedge u \neq v \rightarrow \lvert X(u) \triangle X(v) \rvert = 1 \wedge$ $\forall x \in X(u) \triangle X(v).exposedTo(x) = (determines \circ of)(c)$ (X(u): the exposures of the treatment of u, as substance, dose and time) *beyond SROIQ: compares two individuals' sets of exposures (rule with aggregation, SPARQL, or the checker)*
`rescue-after-challenge`	ALC(D) feature chains	$\exists hasRole$.RescuedRole $\sqsubseteq$ $\exists(rescueExposure \circ atTime, challengeExposure \circ atTime). >$ *with rescueExposure, challengeExposure functional subroles of hasTreatment∘hasExposure; not OWL 2*

7 The challenge-rescue experiment as an instance

The ABox below is the experiment of `experiment.ldl` as derived (phases and steps from the procedure) and placed (plate and wells from the mapping to Cell 01). Doses and times are omitted.



ChallengeRescueExperiment(e)
hasQuestion(e, q)
RecoveryModel(M1_graded)
hasCandidate(q, M1_graded)
RecoveryModel(M2_threshold)
hasCandidate(q, M2_threshold)
RecoveryModel(M3_protective)
hasCandidate(q, M3_protective)
Belief(b)
hasBelief(e, b)
Substance(s_challenge)
hasRole(s_challenge, challengeRole)
hasSubstance(e, s_challenge)
Substance(s_rescue)
hasRole(s_rescue, rescueRole)
hasSubstance(e, s_rescue)
Culture(rescued)
hasCulture(e, rescued)
hasRole(rescued, rescuedRole)
Treatment(t_rescued)
hasTreatment(rescued, t_rescued)
Exposure(x_rescued_challenge)
hasExposure(t_rescued, x_rescued_challenge)
exposedTo(x_rescued_challenge, s_challenge)
Exposure(x_rescued_rescue)
hasExposure(t_rescued, x_rescued_rescue)
exposedTo(x_rescued_rescue, s_rescue)
Culture(comparator)

hasCulture(e, comparator)
hasRole(comparator, comparatorRole)
Treatment(t_comparator)
hasTreatment(comparator, t_comparator)
Exposure(x_comparator_challenge)
hasExposure(t_comparator, x_comparator_challenge)
exposedTo(x_comparator_challenge, s_challenge)
Culture(control)
hasCulture(e, control)
hasRole(control, controlRole)
EmptyTreatment(t_control)
hasTreatment(control, t_control)
Comparison(c_validity)
hasComparison(e, c_validity)
between(c_validity, comparator)
between(c_validity, control)
licenses(c_validity, c_primary)
Comparison(c_primary)
hasComparison(e, c_primary)
between(c_primary, rescued)
between(c_primary, comparator)
Insight(ins_answer)
hasInsight(e, ins_answer)
from(ins_answer, c_primary)
Insight(ins_effect_rescue)
hasInsight(e, ins_effect_rescue)
from(ins_effect_rescue, c_primary)
Insight(ins_effect_challenge)
hasInsight(e, ins_effect_challenge)
from(ins_effect_challenge, c_validity)
CultivationPhase(cultivation)
hasPhase(e, cultivation)
InterventionAction(x1)
hasStep(cultivation, x1)
InterventionAction(x2)
hasStep(cultivation, x2)
GrowthAction(g1)
hasStep(cultivation, g1)
InterventionAction(x3)
hasStep(cultivation, x3)
GrowthAction(g2)
hasStep(cultivation, g2)
ObservationPhase(observation)
hasPhase(e, observation)
ObservationAction(o1)
hasStep(observation, o1)
ObservationAction(o2)
hasStep(observation, o2)
ObservationAction(o3)
hasStep(observation, o3)
AnalysisPhase(analysis)
hasPhase(e, analysis)
InterpretationAction(i1)
hasStep(analysis, i1)
InterpretationAction(i2)
hasStep(analysis, i2)
InterpretationAction(i3)
hasStep(analysis, i3)
AnalysisAction(v1)
hasStep(analysis, v1)
AnalysisAction(a1)
hasStep(analysis, a1)
realizes(x1, x_rescued_challenge)
realizes(x2, x_comparator_challenge)
realizes(x3, x_rescued_rescue)
realizes(v1, c_validity)
realizes(a1, c_primary)
Plate(P17)
hasSite(e, P17)
Well(A01)
hasWell(P17, A01)
contains(A01, rescued)
Well(A02)
hasWell(P17, A02)
contains(A02, comparator)
Well(A05)
hasWell(P17, A05)
contains(A05, control)

Conformance and classification (closed world)

concept	constraints	satisfied
Experiment	4	4
ComparativeExperiment	17	17
ControlledExperiment	3	3
HypothesisDiscriminatingExperiment	5	5
ChallengeRescueExperiment	7	7

Most specific concept: ChallengeRescueExperiment (declared: ChallengeRescueExperiment). Every part satisfies its own concept (50/50).

Negative controls

modification	fails at	still an instance of
comparator dose 0.60 au	ControlledExperiment	HypothesisDiscriminatingExperiment, ComparativeExperiment
control culture removed	ComparativeExperiment, ControlledExperiment	HypothesisDiscriminatingExperiment
rescue given with the challenge (0 h)	ChallengeRescueExperiment	HypothesisDiscriminatingExperiment, ControlledExperiment
belief and answer removed	HypothesisDiscriminatingExperiment	ControlledExperiment
rescue intervention missing from the plan	ComparativeExperiment	HypothesisDiscriminatingExperiment
comparator on another plate	ComparativeExperiment	HypothesisDiscriminatingExperiment
an observation step in two phases	ComparativeExperiment	HypothesisDiscriminatingExperiment
a well as part of a treatment	Experiment, ComparativeExperiment	—